



UNIVERSITY OF TRIESTE

Department of Mathematics, Informatics and Geosciences

Bachelor's Degree in Artificial Intelligence and Data Analytics

BACHELOR'S THESIS

**Cross-View Association of Identical Parts Across
Non-Overlapping 3D Scans: Identifiability,
Shift Ambiguity and Evaluation**

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Chapter 1

Introduction

1.1 The problem

Industrial handling is increasingly automated. Repetitive and heavy tasks such as lifting or sorting parts, once assigned to human operators, are now delegated to robots to satisfy the cycle time required by the production line.

For a robot to grasp an object, it is necessary to determine the object's pose, which dedicated sensors measure as a position and an orientation relative to the robot's own coordinate frame.

For a single compact piece, one 3D sensor is usually sufficient, as its viewpoint can cover the entire object. On the other hand, a very long part cannot be captured from a single viewpoint, making a single scanner insufficient. The layout therefore requires two sensors, one for each end. The part is gripped precisely at these ends, whose faces also determine the 6D pose of a straight extruded profile up to the symmetry of the cross-section.

The setup consists of two structured-light 3D scanners, positioned at either end of a conveyor carrying bundles of six-metre metal bars of identical profile.

Each device is mounted on a dedicated carriage at one end of the bundle, facing the end faces of the bars rather than their length. The two carriages are separated by several metres along the axis of the bar. The part of the bundle that each scanner covers is often narrower than the bundle itself, so a single view does not always contain every bar. Because the two sensors face opposite ends, their respective viewpoints neither overlap nor capture any common surface. Finally, a dual-arm manipulator picks up each bar by gripping it at both ends. Figure 1.1 illustrates this setup.

The task is therefore to determine which detection from one scanner and which from the other belong to the same physical object. In this specific case, data association cannot be resolved using conventional techniques due to the absence of typical geometric or visual features.

Absence of epipolar constraint. The lack of overlap between the two views means that the scanners never see the same surface.

Visual appearance invariance. Because the bars originate from the same extrusion batch, end faces belonging to the same profile family (i.e., bars sharing the same cross-section) are identical up to saw marks.

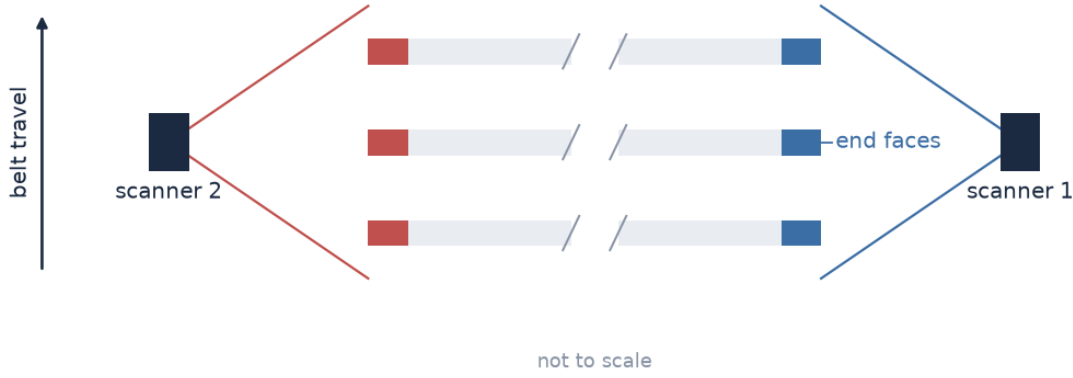


Figure 1.1: Dual-scanner system. Top view of the system. Each scanner is positioned at one end of the bundle and detects only the end faces of the bars facing it. In the figure, these faces are highlighted at both ends of each bar. The viewpoints of the two scanners do not share any surface and the bundle passes through both. Neither axis is to scale.

Relative position invariance. The bars are packed edge to edge at an almost uniform pitch, so the scanner detects a sequence of end faces that remains visually invariant even if the bundle shifts by one position.

Conveyor motion between acquisitions. Because the conveyor belt advances between the two acquisitions, the two scan windows can cut the bundle at different bars, in which case an association that pairs the detections in their order is shifted by one or more bars.

This shift ambiguity is illustrated in fig. 1.2. In each view, a different end bar remains excluded because each view often covers less than the whole bundle and the bundle moved between the two acquisitions. A correct association should record two unpaired detections, whereas a complete sequential association would lead to a systematic error.

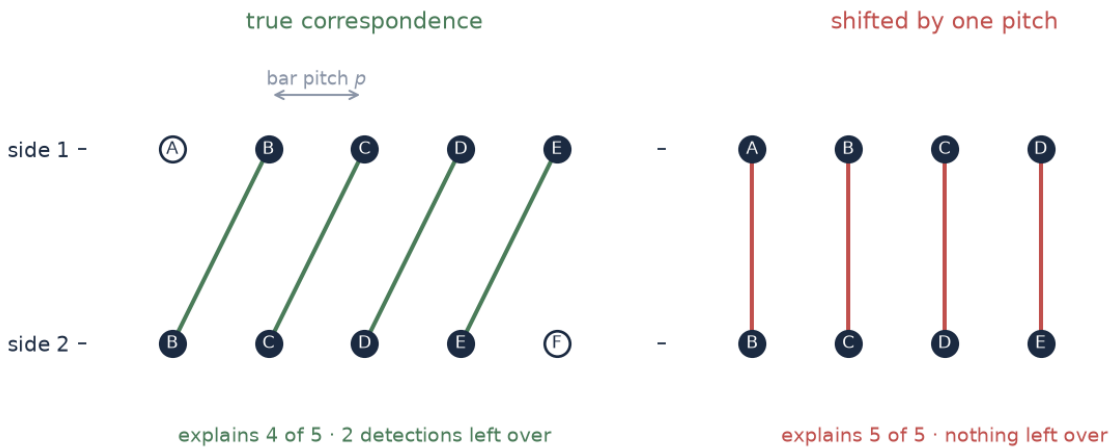


Figure 1.2: The shift ambiguity (schematic). Each scanner covers a different part of the bundle, so a different bar is truncated in each acquisition. A correspondence shifted by one pitch would incorrectly pair all detections, whereas the correct correspondence leaves the two truncated detections unpaired. Letters identify the physical bar and hollow circles mark the unpaired detections.

This condition is not an edge case because in many scenes of the dataset some bars are

scanned at only one end. The objective is to determine what portion of the correspondence is actually recoverable and to define how unpaired detections are handled.

The scans used in this thesis were provided by beanTech, the company that developed the vision software of the rig for the manufacturer that operates it. At the time of this work the rig was still being commissioned and only a limited number of scenes was available. This setting favours methods that require no training data, such as the classical detector and the pairing methods studied here, since the detector needs only the drawing of the bar cross-section.

